BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: —

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The vesicles for the formation of phragmoplast originate from:

(A) Endoplasmic reticulum
(B) Golgi apparatus
(C) Nucleus
(D) Mitochondria

Q2. The percentage of unsaturated fatty acids in sunflower oil is:

(A) 70%
(B) 76%
(C) 80%
(D) 85%

Q3. Bacteria are assigned to which the kingdom:

(A) Fungi
(B) Monera
(C) Protista
(D) Animalia

Q4. To which group of molecules enzymes belong to:

(A) Carbohydrates
(B) Proteins
(C) Lipids
(D) Nucleic Acids

Q5. Narrowing of arteries is called:

(A) Plaques
(B) Atherosclerosis
(C) Arteriosclerosis
(D) Embolus

Q6. The study of internal structures of living things is called:

(A) Histology (B) Morphology

(C) Anatomy (D) Microbiology

Q7. Prokaryotes have a cell wall composed of a chemical called:

(A) Lipids (B) Chitin

(C) Peptidoglycan (D) Cellulose

Q8. Group of organisms capable of inter-breeding and producing fertile offspring is called:

(A) Ecosystem (B) Population

(C) Community (D) Species

Q9. In which stage of meiosis crossing over takes place:

(A) Prophase-I (B) Metaphase-I

(C) Anaphase-I (D) Telophase-I

Q10. Which one of these organelles has its own DNA:

(A) Chloroplast (B) Ribosome

(C) Mitochondria (D) Golgi body

Q11. Biological method has been playing an important role in scientific research for the last:

(A) 400 years (B) 450 years

(C) 500 years (D) 550 years

Q12. In photosynthesis chlorophyll absorb light having colours:

(A) Green and blue (B) Blue and red

(C) Orange and blue (D) Green and orange

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Define Biology.

(ii) What is zone of life?

(iii) Write scientific name of mustard plant and frog.

(iv) What is biological problem?

(v) Define observations and write their types.

(vi) Write the two observations of AFA King.

(vii) Define magnification and resolution power.

(viii) What is hypertonic solution and hypotonic solution?

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Differentiate between taxonomy and systematics.

(ii) Define species.

(iii) Differentiate between chiasmata and crossing over.

(iv) What is difference between malignant and benign tumors?

(v) Define cell cycle.

(vi) Define chromosomes.

(vii) What do you mean by photosystem.

(viii) Define ATP.

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) Define inhibitors and activators.

(ii) Define biocatalysts and write two benefits of biocatalysts.

(iii) Define the specificity of enzyme. Give an example.

(iv) What is meant by Villi? Write down its function.

(v) Give two symptoms of constipation.

(vi) Differentiate between source and sink.

(vii) Define angioplasty and bypass surgery.

(viii) Differentiate between atherosclerosis and arteriosclerosis.

> Note: Attempt any TWO questions. (PART-II)

Q5.

(a) [5] Explain relationship of biology with five other branches of science.

(b) [4] Write a note on endocytosis Explain your answer with labelled diagram.

Q6.

(a) [5] Define the term "Extinct" and explain about endangered species of Pakistan.

(b) [4] Write a delated note on electron transport chain.

Q7.

(a) [5] Describe the digestion of food in small intestine.

(b) [4] Explain the composition and functions of white blood cells.

GROUP – II

Objective Type

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Branch of biology which deals with study of parasites is called:

(A) Ecology
(B) Morphology
(C) Genetics
(D) Parasitology

Q2. Career which deals with the art of gardening is called:

(A) Medicine
(B) Agriculture
(C) Forestry
(D) Horticulture

Q3. The freezing point of water is:

(A) 0°C
(B) 37°C
(C) 50°C
(D) 100°C

Q4. Mushrooms belong to kingdom:

(A) Plantae
(B) Protista
(C) Fungi
(D) Monera

Q5. Water and dissolved substances are transported by:

(A) Xylem
(B) Phloem
(C) Stomata
(D) Leaves

Q6. A major component of plant cell wall is:

(A) Chitin
(B) Cellulose
(C) Cholesterol
(D) Glucose

Q7. Meiosis does not occur in kingdom:

(A) Monera
(B) Fungi
(C) Plantae
(D) Animalia

Q8. Phase of mitosis in which chromosomes arrive at the poles is:

(A) Anaphase
(B) Metaphase
(C) Telophase
(D) Prophase

Q9. Protease acts only on:

(A) Acids
(B) Carbohydrates
(C) Protein
(D) Lipids

Q10. Dark reactions were discovered by:

(A) Malvin Calvin
(B) Karl Lohmann
(C) Koshland
(D) T. Schwann

Q11. One gram of lipid contains energy:

(A) 4 k. cal
(B) 9 k. cal
(C) 10 k. cal
(D) 12 k cal

Q12. Blood vessels that carry blood towards heart are called:

(A) Veins
(B) Arteries
(C) Capillaries
(D) Xylem

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Define bioeconomics.

(ii) State about Bu Ali Sina achievements in Biology.

(iii) What is biosphere level.

(iv) Define scientific method.

(v) Write two characteristics of hypothesis.

(vi) Define incubation period.

(vii) Differentiate between sieve tubes cells and companion cells.

(viii) Define active transport.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Define systematics.

(ii) Write two aims of classification.

(iii) Differentiate between karyokinesis and cytokinesis.

(iv) How phragmoplast is formed in plant cell?

(v) How mitosis helps in cell replacement of body?

(vi) What is meant by tetrad?

(vii) How photosynthesis gets effected by temperature?

(viii) Why it is incorrect to say that the energy releasing step of respiration is electron transport chain?

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) How enzymes are used as biological detergent?

(ii) What does saturation of active sites mean?

(iii) Define "Lock & Key Model" of enzymes.

(iv) Why vitamin C is important?

(v) Define peristalsis.

(vi) What is "Cohesion Tension Theory"?

(vii) Define arteries and veins.

(viii) Write the effects of dengue fever on platelets.

> Note: Attempt any TWO questions. (PART-II)

Q5.

(a) [5] Define biology and explain its three main divisions.

(b) [4] Write a note on Golgi apparatus and Lysosome.

Q6.

(a) [5] Define endangered species and discuss any three types of endangered species in Pakistan.

(b) [4] Write a note on aerobic and anaerobic respiration.

Q7.

(a) [5] Describe complete digestion in small intestine.

(b) [4] Write a detailed note on blood plasma.